

Data Mining in Hive – Exam Ready Notes

1. Table Creation

```
CREATE TABLE student (  
  name STRING,  
  marks INT,  
  roll_no INT  
)  
ROW FORMAT DELIMITED  
FIELDS TERMINATED BY ','  
STORED AS TEXTFILE;
```

2. Load Data

```
LOAD DATA LOCAL INPATH '/path/student_data.csv'  
INTO TABLE student;
```

3. Important Queries

Top Performers

```
SELECT name, marks FROM student  
ORDER BY marks DESC LIMIT 5;
```

Average Marks

```
SELECT AVG(marks) FROM student;
```

Grade Classification

```
SELECT name, marks,  
CASE  
  WHEN marks >= 75 THEN 'Distinction'  
  WHEN marks >= 60 THEN 'First Class'  
  WHEN marks >= 50 THEN 'Second Class'  
  ELSE 'Fail'  
END AS grade  
FROM student;
```

Students Above Average

```
SELECT name, marks FROM student  
WHERE marks > (SELECT AVG(marks) FROM student);
```

Ranking Students

```
SELECT name, marks,  
RANK() OVER (ORDER BY marks DESC) AS rank
```

```
FROM student;
```

Low Performers

```
SELECT name, marks FROM student  
WHERE marks < 40;
```

Marks Classification

```
SELECT  
CASE  
  WHEN marks BETWEEN 0 AND 40 THEN 'Low'  
  WHEN marks BETWEEN 41 AND 70 THEN 'Medium'  
  ELSE 'High'  
END AS category,  
COUNT(*)  
FROM student  
GROUP BY category;
```

4. Key Concepts

- Aggregation: AVG, COUNT - Classification: CASE - Ranking: RANK() - Filtering: WHERE - Grouping: GROUP BY